

NeoNegotiator: A Human-Aware Automated Negotiating Agent for the ANAC 2026 HAN League

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Abstract

This report presents the final version of **NeoNegotiator**, an automated negotiating agent submitted to the Human-Agent Negotiation (HAN) league of ANAC 2026. NeoNegotiator extends `LLMMetaNegotiator` with a two-phase architecture: an *interrogation phase* that observes the opponent’s bidding curve for up to 15 rounds, followed by a *counter phase* that selects one of 13 purpose-built bidding strategies from a 260-entry counter-strategy table indexed by opponent character (Thomas–Kilman), opponent emotion, and detected strategy type. Opponent character is derived from bid-delta analysis; emotion is detected per round via LLM sentiment classification. Six human-targeted overlays (issue logrolling, sentiment-adaptive tone, door-in-the-face anchoring, BATNA signaling, deadline-delay pressure, and micro-step concessions) are independently toggleable. Automated benchmarks across three HAN scenarios achieve a 100% agreement rate against all classical baselines, with issue logrolling providing a consistent +1.0% mean-advantage gain and auto-counter achieving 79.5% opponent-strategy detection accuracy. The agent is submitted to ANAC 2026 as `neonegotiator.NeoNegotiator`.

Keywords: automated negotiation, human-agent negotiation, opponent modeling, Thomas–Kilman model, strategy detection, large language models, logrolling, counter-strategy

1 Introduction

The ANAC 2026 Human-Agent Negotiation (HAN) league challenges agents to negotiate bilaterally with human subjects using the Alternating Offers Protocol (AOP) augmented with natural-language text [1, 8]. Unlike purely automated leagues, HAN performance depends on both rational bid selection and *persuasion*: generating contextually appropriate messages that influence a human opponent’s acceptance decisions.

This report traces the design of **NeoNegotiator** from its Progress-Report prototype to its final submitted form. The original prototype implemented four novelty components over a single Boulware core. Through iterative experimentation and testing, the bidding core was expanded to 13 strategies with automated opponent-strategy detection, a 260-

entry counter-strategy table, and a rolling re-analysis mechanism.

Section 2 reviews the negotiation framework. Section 3 documents the design evolution and problems encountered. Section 4 describes the final agent in full. Section 5 covers implementation. Section 6 presents experiments. Section 7 interprets the findings. Section 8 concludes.

2 Background

2.1 Alternating Offers Protocol

At each round $t \in \{1, \dots, T\}$ an agent either proposes an outcome $\omega \in \Omega$ or responds ACCEPT, REJECT, or ENDNEGOTIATION [5]. Each agent i has a private additive utility function $u_i(\omega) = \sum_k w_k \cdot v_k(\omega_k)$ and a reserved value r_i .

2.2 Time-Based Bidding

Aspiration-based strategies target

$$\alpha(t) = r + (\alpha_{\max} - r)(1 - (t/T)^\beta), \quad (1)$$

where $\beta > 1$ is Boulware (concedes late) and $\beta < 1$ is Conceder (concedes early) [5].

2.3 Frequency-Based Opponent Modeling

Frequency models estimate the opponent’s preferences from observed offers by counting how often each issue value appears [13]. The estimated opponent utility is a normalised weighted count (see Equation 3). NeoNegotiator uses this model for logrolling, Bayesian, KDE, Trade-Off, and Pareto strategies.

2.4 Opponent Behavior Modeling

Beyond preference modeling, the agent must infer the opponent’s *negotiation behaviour*—their concession pattern over time [9]. NeoNegotiator classifies opponent behaviour into one of 13 standard strategy types (Boulware, Conceder, Tit-for-Tat, etc.) and uses this to select a counter-strategy.

2.5 HAN League Specifics

Agents must use the locally hosted `qwen3:4b-instruct` model via Ollama [6]; only hyperparameters such as temperature are configurable.

Agents are ranked by utility achieved and human-perceived fairness via questionnaires [1].

3 Design Evolution

3.1 Original Plan (Progress Report)

The Progress Report prototype targeted four novelty components:

1. **Issue Logrolling** — find outcomes at the same utility level that are better for the opponent, producing Pareto improvements [11].
2. **Opponent Character + Emotion Profiling** — two-layer model: persistent character from bid-delta analysis, dynamic emotion (Frustrated / Happy / Anxious / Neutral) from LLM sentiment; the pair drives LLM tone via a Tone Matrix.
3. **Door-in-the-Face Anchoring** — open with maximum utility, then frame the first concession as a generous gesture in the LLM text.
4. **BATNA Signaling** — inject a walk-away hint when the opponent is stubborn, time exceeds 60%, and fewer than two signals have been sent [7].

The planned bidding core was a single Boulware curve switching to Conceder at 65% elapsed time when the opponent appeared unresponsive.

3.2 Problems Encountered and Resolutions

Problem 1: Single-strategy core is too rigid. A fixed Boulware curve reaches agreement too early against Conceder opponents (low utility point) and stalls against Tit-for-Tat opponents who mirror its moves. *Resolution:* the bidding core was expanded to 13 independently implemented strategies and an auto-counter mechanism was added that detects the opponent’s strategy type during an initial interrogation phase and switches to a purpose-selected counter strategy [3, 14].

Problem 2: Four-class character model is too coarse. The original Stubborn / Cooperative / Aggressive / Rational taxonomy produced ambiguous classifications for opponents with moderate concession rates and low cooperation. *Resolution:* the character model was replaced with the Thomas–Kilman five-style taxonomy (Competing, Collaborating, Compromising, Avoiding, Accommodating) [12], which is grounded in two orthogonal axes (assertiveness, cooperativeness) and provides cleaner decision boundaries.

Problem 3: Island scenario utility scale breaks the classifier. The Island scenario uses unnormalized utilities in $[0, 100]$ rather than $[0, 1]$, causing per-step deltas to be $100\times$ larger than expected thresholds. *Resolution:* auto-detection of scale; if $\max(\text{observed utilities}) > 1.5$, all values are divided by 100 before threshold comparisons.

Problem 4: Tft opponents contaminate re-classification. After the counter strategy is activated, Tit-for-Tat opponents begin mirroring the agent’s new bidding curve, causing the classifier to detect a different strategy on the accumulated history and trigger spurious counter-switches. *Resolution:* a

Algorithm 1: NeoNegotiator high-level control flow

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1: Phase 1 — Interrogation (rounds 0
   ...interrogate_rounds)
2:   Use Boulware bids; collect opponent offer utilities
3:   At round probe_round: deliberate concession
   step; record mirror response
4:   If observations  $\geq$  interrogate_rounds: go to
   Phase 2
5: Phase 2 — Counter
6:   CLASSIFY(offer history)  $\rightarrow$  detected strategy type
7:   counter  $\leftarrow$  CounterTable[character, emotion, de-
   tected]
8:   Switch active bidding strategy to counter
9:   Every re_analyze_interval observations: re-
   classify (two-match rule)
10: Each round (both phases):
11:   Apply logrolling to candidate outcome
12:   Call LLM with tone = ToneMatrix[character, emo-
   tion]
13:   Return ExtendedOutcome(outcome, text)

```

two-consecutive-match rule — a strategy switch only fires if the same new type is detected on two successive re-analysis cycles 15 rounds apart.

Problem 5: LLM calls make tournaments prohibitively slow. A 30-round session with both parties calling the LLM takes 3–5 minutes, making full ablation experiments infeasible. *Resolution:* a skip_llm flag disables all LLM calls and returns deterministic fallback text, enabling fast automated benchmarking.

3.3 Additions Beyond the Original Plan

Three capabilities were added that were not in the original design:

- **Active probe:** at round 8 the agent makes a deliberate 5% concession step to test whether the opponent mirrors it, enabling Tit-for-Tat detection before the standard 15-round interrogation completes [3].
- **Rolling re-analysis:** after the counter phase begins, the classifier re-runs on the full accumulated history every 15 new observations, allowing mid-negotiation strategy adaptation.
- **KDE and Pareto strategies:** Kernel Density Estimation of the opponent’s acceptance zone and Pareto-frontier navigation were added as high-information counter-strategies effective after sufficient offer history is collected.

4 Final Agent Design

4.1 Architecture Overview

NeoNegotiator inherits from LLMMetaNegotiator and operates in two sequential phases per negotiation session (Algorithm 1).

Table 1: NeoNegotiator bidding strategies.

Strategy	Description
Boulware (BOu)	Polynomial $\beta=5$; concedes sharply near deadline [5]
Conceder (CON)	Polynomial $\beta=0.2$; concedes fast then flattens [5]
Linear (LIN)	Uniform decline from u_{max} to r
TDT	Generalised time-dependent tactic; configurable β
Tit-for-Tat (TFT)	Mirrors opponent's last absolute concession [3]
Relative TFT (RTF)	Mirrors opponent's relative concession rate [3]
Nice TFT (NTF)	Opens 5% below max to signal goodwill; then mirrors [3]
Deadline-Driven (DDR)	Holds max until switch point; then concedes sharply
Random Walk (RWK)	Gaussian noise around a linear baseline
Bayesian (BAY)	Maximises $u_{neo} \times \hat{p}(\text{opp accepts})$ under aspiration floor
KDE	Kernel density estimate of opponent's acceptance zone
Trade-Off (TRO)	$\alpha u_{neo} + (1-\alpha) \hat{u}_{opp}$; α decreases over time
Pareto (PAR)	Navigates Pareto frontier; selects outcome closest to aspiration

4.2 Bidding Strategies

Thirteen bidding strategies are available, each implementing a common `propose(state)` interface (Table 1).

4.3 Opponent Strategy Classifier

The classifier computes six signals from the sequence of utilities received from the opponent's offers [9]:

- $\bar{\delta}$ — mean per-step delta (overall concession rate).
- Δ_{accel} — second-half minus first-half rate (acceleration).
- $\sigma_{\bar{\delta}}^2$ — variance of per-step deltas.
- $\bar{\delta}_{\text{first}}, \bar{\delta}_{\text{last}}$ — rates in first and last third.
- $\rho_{\bar{\delta}}$ — lag-1 autocorrelation of deltas (negative = oscillating).
- f_{mono} — fraction of steps where opponent improved toward us.

All rates are normalised to a canonical 30-step game ($\times 30/T$) so thresholds are scenario-independent. A priority-ordered rule set maps signal combinations to strategy types; the active probe result (mirroring detected at round 8) resolves Boulware / TtT ambiguity in the near-zero-movement region [3].

4.4 Counter-Strategy Table

The counter-strategy table (Table 2) has $5 \times 4 \times 13 = 260$ entries keyed by Thomas–Kilman character, opponent emotion, and detected opponent strategy. Design principles:

- *Competing* opponents: Boulware or Deadline-Driven to avoid exploitation.
- *Collaborating*/Happy opponents: Pareto to maximise joint utility.
- *Avoiding* opponents: Deadline-Driven to force a deal before disengagement.
- *Accommodating* opponents: Boulware dominates; they concede freely.
- Anxious emotion $\rightarrow$ Linear (predictable, non-threatening).
- Frustrated emotion $\rightarrow$ firm counter (Boulware / TtT).

Table 2: Complete counter-strategy table ($5 \times 4 \times 13 = 260$ entries). Row: character $\times$ emotion (FRU=Frustrated, HAP=Happy, ANX=Anxious, NEU=Neutral). Column: detected opponent strategy. Cell: our counter-strategy. Strategy abbreviations in Table 1.

Char.	Emo	Detected Opponent Strategy												
		BOu	CON	LIN	TDT	TFT	RTF	NTF	DDR	RWK	BAY	KDE	TRO	PAR
Competing	FRU	DDR	BOu	BOu	BOu	BOu	BOu	BOu	DDR	KDE	RWK	RWK	BOu	BOu
	HAP	NTF	BOu	BOu	NTF	NTF	NTF	TRO	LIN	KDE	RWK	RWK	PAR	BAY
	ANX	DDR	BOu	BOu	DDR	BOu	BOu	DDR	LIN	KDE	RWK	RWK	DDR	DDR
	NEU	DDR	BOu	BOu	BAY	NTF	BOu	BOu	LIN	KDE	RWK	RWK	PAR	BAY
Collab.	FRU	NTF	TRO	NTF	NTF	NTF	TRO	LIN	KDE	TRO	TRO	PAR	PAR	BAY
	HAP	PAR	PAR	PAR	PAR	PAR	PAR	PAR	KDE	PAR	PAR	PAR	PAR	PAR
	ANX	LIN	LIN	LIN	LIN	LIN	LIN	LIN	KDE	LIN	LIN	LIN	LIN	LIN
	NEU	BAY	PAR	PAR	BAY	NTF	PAR	PAR	LIN	KDE	RWK	TRO	PAR	BAY
Comp.	FRU	TFT	BOu	BOu	TFT	NTF	TFT	TFT	LIN	KDE	RWK	RWK	PAR	BAY
	HAP	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu
	ANX	LIN	LIN	LIN	LIN	LIN	LIN	LIN	LIN	KDE	LIN	LIN	LIN	LIN
	NEU	DDR	BOu	BOu	BAY	TDT	BOu	BOu	LIN	KDE	KDE	RWK	PAR	BAY
Avoiding	FRU	CON	CON	CON	CON	NTF	CON	TRO	LIN	CON	CON	CON	CON	TRO
	HAP	TRO	TRO	NTF	TRO	NTF	NTF	TRO	LIN	KDE	TRO	TRO	PAR	TRO
	ANX	CON	LIN	LIN	LIN	CON	CON	LIN	LIN	CON	CON	CON	LIN	CON
	NEU	DDR	DDR	DDR	DDR	DDR	DDR	TRO	LIN	KDE	DDR	DDR	DDR	DDR
Accom.	FRU	NTF	NTF	NTF	NTF	NTF	NTF	TRO	LIN	KDE	NTF	NTF	TRO	NTF
	HAP	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu
	ANX	BOu	BOu	BOu	BOu	BOu	BOu	BOu	LIN	KDE	BOu	BOu	BOu	BOu
	NEU	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	BOu	TRO

4.5 Thomas–Kilman Character Model

Character is classified from the 5-round rolling utility stream of the opponent's offers [12]:

$$\text{rate} = \frac{1}{n-1} \sum_i (u_{i+1} - u_i), \quad \text{level} = u_{\text{last}}. \quad (2)$$

Rules (applied in priority order): $\text{rate} < -0.01 \Rightarrow \text{Competing}$; $\text{rate} > 0.04 \Rightarrow \text{Accommodating}$; $|\text{rate}| < 0.005 \wedge \text{low variance} \Rightarrow \text{Avoiding}$; $\text{rate} > 0.01 \wedge \text{level} > 0.4 \Rightarrow \text{Collaborating}$; otherwise *Compromising*.

4.6 Opponent Preference Model

A per-issue value-frequency table is maintained: $C_{i,v}$ counts how often the opponent proposed value v on issue i [13]. The estimated opponent utility is:

$$\hat{u}_{\text{opp}}(\omega) = \frac{1}{n} \sum_{i=1}^n \frac{C_{i,\omega_i}}{\max_v C_{i,v}}. \quad (3)$$

4.7 Acceptance Strategy

Acceptance delegates to `BoulwareTBNegotiator.respond()`, which implements the ACNext criterion: accept iff $u_{\text{neo}}(\omega_{\text{opp}}) \geq u_{\text{neo}}(\omega_{i+1}^*)$, i.e. the opponent's offer is at least as good as what the agent would propose next [2]. The Deadline-Delay overlay overrides this to block acceptance before the switch point.

4.8 Issue Logrolling

After the bidding strategy selects ω^* with utility u^* , logrolling searches the outcome space for ω' with:

$$|u_{\text{neo}}(\omega') - u^*| \leq 0.025 \quad \text{and} \quad \hat{u}_{\text{opp}}(\omega') > \hat{u}_{\text{opp}}(\omega^*). \quad (4)$$

The best such ω' replaces ω^* , producing a Pareto-improving trade without sacrificing own utility [11].

Table 3: Tone Matrix: representative entries.

Character	Emotion	Tone instruction
Competing	Frustrated	Stay firm; acknowledge friction without yielding; reframe as mutual risk
Competing	Anxious	Signal no deadline pressure; let the tension work for you
Collaborating	Happy	Build positive momentum; propose a value-creating trade to close
Collaborating	Anxious	Reassure progress is real; suggest a specific path to agreement
Compromising	Frustrated	Show movement; a visible step toward middle ground rebuilds trust
Avoiding	Neutral	Short and direct; make it easy for them to respond
Accommodating	Frustrated	Express appreciation for flexibility; hold remaining position firmly
Accommodating	Happy	Warm but decisive; do not let goodwill tempt over-concession

Table 4: Key NeoNegotiator constructor parameters.

Parameter	Default	Effect
auto_counter	True	Interrogation + counter-switch
interrogate_rounds	15	Offers to observe before switching
probe_round	8	Round for active TFT probe
re_analyze_interval	15	Counter-phase re-check interval (obs.)
use_logrolling	True	Pareto-improving issue trades
use_sentiment	False	LLM-based emotion classification
use_door_in_face	False	Round-0 maximum-utility anchor
use_batna	False	Walk-away hint vs. stubborn opponents
use_deadline_delay	False	Freeze bids for first 65% of deadline
use_micro_steps	False	Linear concession curve from round 10
skip_llm	False	Disable LLM calls (fast testing mode)

4.9 LLM Communication and Tone Matrix

The LLM (qwen3:4b-instruct, temperature 0.5) is called once per round and must return a JSON object with two fields: "message" (1–3 persuasive sentences sent to the opponent) and "debug" (3–5 bullet points for the developer, not forwarded). If the LLM call fails or returns malformed JSON, a deterministic fallback message is used.

The system prompt appends the (character, emotion) pair’s tone instruction from the Tone Matrix (Table 3), which maps all 20 opponent states to a personalised communication directive [4, 10].

5 Implementation

5.1 Technology Stack

NegMAS $\geq 0.15.2$ (negotiation platform) [6]; negmas-llm $\geq 0.4.2$ (LLMMetaNegotiator base class); Ollama with qwen3:4b-instruct (local LLM inference); scipy (optional, enables KDE; graceful fallback to frequency scoring); hani $\geq 0.3.0$ (HAN scenario utilities).

5.2 Key Constructor Parameters

5.3 Submission

The submission zip contains two files: neonegotiator.py (≈ 92 KB) and requirements.txt. The agent is registered on the ANAC portal as module neonegotiator, class NeoNegotiator.

Table 5: Ablation results: 18 sessions per configuration against classical baselines.

Configuration	Mean Adv.	Agree %	Sessions
Baseline (Boulware)	23.22	100%	18
+Logrolling (S1)	23.45	100%	18
+Sentiment (S2)	23.22	100%	18
+Door-in-Face (S3)	23.22	100%	18
+BATNA (S4)	23.22	100%	18
+Deadline Delay (S5)	22.30	100%	18
+Micro-Steps (S6)	21.04	100%	18
All strategies	21.04	100%	18

6 Experiments

6.1 Setup

All automated experiments use skip_llm=True. Three HAN scenarios: Grocery (3 issues, [0, 1]), Trade (4 issues, [0, 1]), Island (5 issues, [0, 100]). Baselines: BoulwareTBNegotiator, ConcederTBNegotiator, NaiveTitForTatNegotiator, SimpleNeg, BOANeg, TemplateBasedAdapterNegotiator. Each configuration runs in both proposer roles across all three scenarios. Primary metric: **mean advantage** = $u_{\text{neo}}(\omega^*) - r_{\text{neo}}$ (Island’s [0, 100] scale dominates the raw value).

6.2 Strategy Ablation

All configurations achieve 100% agreement rate. Logrolling (S1) gains +1.0% by finding Pareto-improving outcome swaps (Equation 4) even against rational bots [11]. Sentiment (S2), door-in-face (S3), and BATNA (S4) show no measurable effect against scripted classical agents, which is mechanistically expected: these strategies operate through the LLM text channel or through opponent psychology. Deadline delay (S5) and micro-steps (S6) reduce mean advantage by 3.9% and 9.4% respectively; this is discussed in Section 7.

6.3 Auto-Counter Evaluation

NeoNegotiator (auto_counter=True) was tested against fixed-strategy instances (auto_counter=False) across all 13 strategy types and 3 scenarios (39 sessions). The classifier correctly detected the opponent strategy in 31 of 39 sessions (79.5%). Misclassifications concentrated on Tit-for-Tat variants, which mirror the agent’s Boulware probe and produce utilities indistinguishable from Boulware in short windows [3]. In all 39 sessions the adaptive agent matched or outperformed the best fixed-strategy baseline, confirming that auto-counter provides a net benefit even with imperfect detection.

6.4 Competition Status

NeoNegotiator was submitted to ANAC 2026 and completed the LLM opponent evaluation round. Final human-subject round results will be available after IJCAI 2026 (August 2026).

7 Discussion

7.1 Bot–Human Trade-Off in Strategy Design

The negative ablation results for S5 and S6 against classical bots are mechanistically expected and constitute evidence of human-specificity rather than failure. Deadline-Delay (S5) freezes bids at maximum utility for 65% of rounds; a rational bot is unaffected, but human opponents experience increasing time pressure and make larger concessions in the final phase [7]. Micro-steps (S6) make early concessions that rational opponents exploit immediately; human opponents perceive the smooth, equal-sized steps as principled and fair, reducing incentive to hold out [4]. This bot–human tension is a known challenge in the HAN literature [10].

7.2 Auto-Counter Limitations

The 79.5% detection accuracy reflects two fundamental constraints. First, many strategy types produce similar short-horizon utility trajectories (Boulware and Deadline-Driven are indistinguishable until the final third). Second, after the counter strategy is activated, the opponent’s utilities change in response, making the accumulated history a mixture of two regimes [15]. Future work could use a sliding-window classifier that resets after each counter switch.

7.3 Logrolling Against Humans

The +1.0% gain against bots comes purely from the Pareto-improvement in the outcome space, since bots do not reciprocate creative issue packaging [11]. Against human opponents who may be influenced by the gesture, the effect is likely larger.

7.4 Strong and Weak Points

Strong points. 100% agreement rate across all configurations and scenarios; principled multi-dimensional opponent model (strategy $\times$ character $\times$ emotion) with a structured counter for each combination; modular, independently toggleable strategies enabling clean ablation; graceful degradation when LLM or sciply is unavailable.

Weak points. Island’s [0, 100] scale dominates aggregate metrics, obscuring effects on the [0, 1] scenarios; no human-subject baseline available to quantify human-specific benefits of S3, S5, S6; emotion classifier accuracy not independently validated; strategy detection degrades for short negotiation horizons.

8 Conclusion

NeoNegotiator evolved from a four-component prototype into a fully adaptive negotiating agent. Its final design comprises a 13-strategy bidding engine, an automated opponent-strategy classifier with an active probe mechanism, a 260-entry counter-strategy table indexed by Thomas–Kilman character and opponent emotion, and six independently toggleable human-targeted behavioral overlays.

Key contributions:

- A two-phase interrogation-then-counter architecture that adapts bidding strategy from observed opponent behaviour [9].

- A Thomas–Kilman character classifier from bid-delta analysis, providing a principled five-style taxonomy [12].
- A Tone Matrix mapping each of 20 opponent states to a personalised LLM tone instruction [4, 10].
- An active probe at round 8 that resolves Boulware / Tft ambiguity before the full interrogation completes [3].
- 100% agreement rate and consistent Pareto-improvement from logrolling in automated benchmarks [11].

The design process confirmed a fundamental tension between bot-optimal and human-optimal strategies: mechanisms such as deadline pressure and concession-shape control are suboptimal against rational agents but expected to benefit against human opponents susceptible to temporal pressure and fairness framing [7]. Empirical validation against real human subjects in the ANAC 2026 final round remains the primary open question.

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